

• DESKTOP-FIRST • BROWSER ONLY • DEMO

SkillLoop

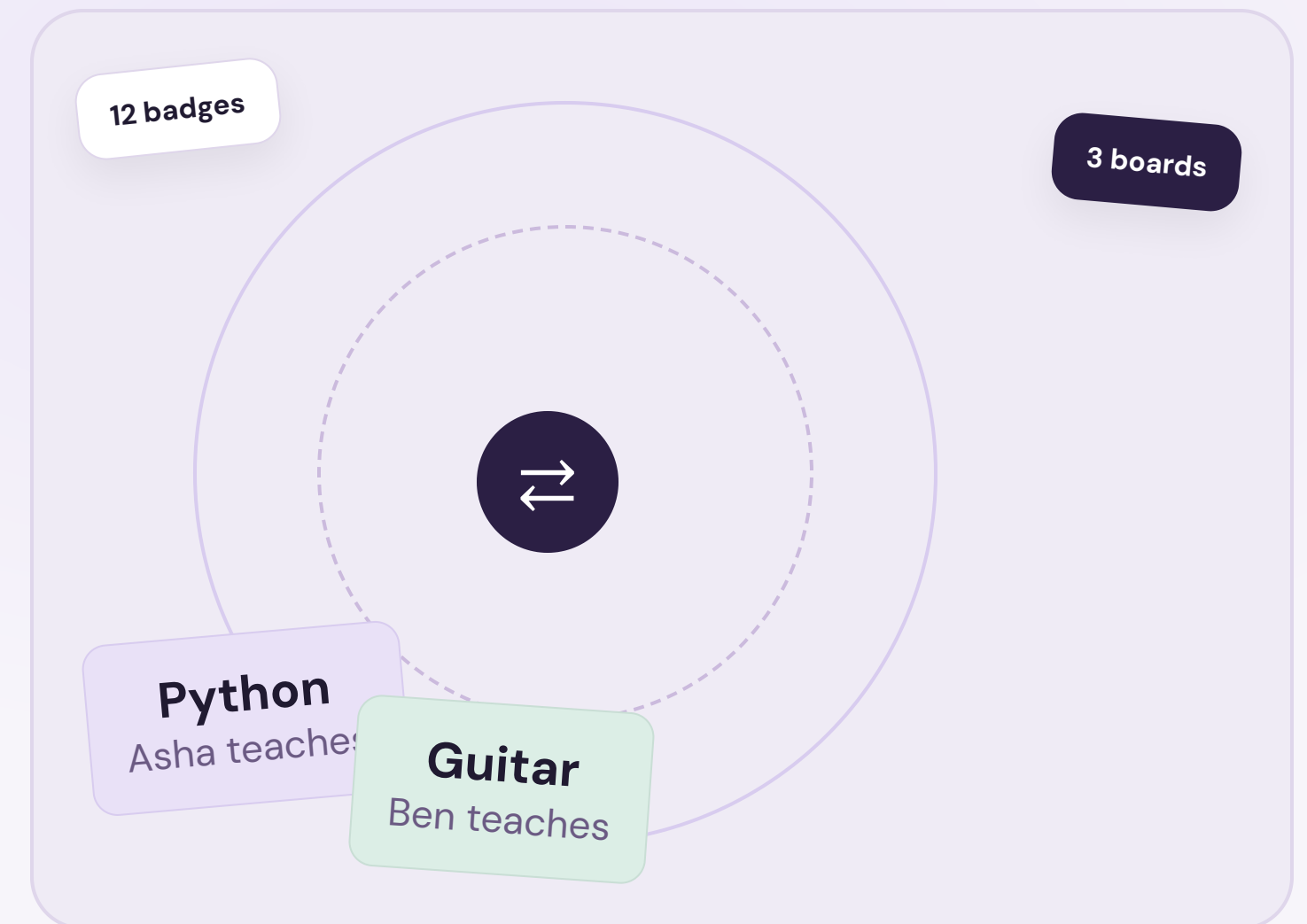
Teach what you know. Learn what you need. Keep a record.

Project rough idea

- One account = learner + teacher in isolated demo workspaces
- Publish a skill with outcome → discover → request → agree on time → learn → record
- 3 ways: free session · direct swap (2 lessons) · paid mentorship (100 demo credits, 15-min hold)
- 3-person Trade Circles when 2-way wants don't align; badges (12) + leaderboard (weekly/monthly/last 7 days)

Team SkillLoop — Hacklith

Add your names / roll nos here. Demo only — no real money, no real identity verification.



02 / PROBLEM STATEMENT

Problem statement

What the site actually addresses — points only.

- Finding the right skill/person is scattered across chats and posts.
- Two-person swaps fail when interests don't align (Asha wants Guitar, Cara wants Python).
- No single place to agree on outcome, language, duration, delivery (online/in-person) and time.
- Attendance alone doesn't show what was learned or built.
- Time-zone and scheduling conflicts cause missed lessons.
- Public posts don't reserve a lesson or create a commitment.

EXAMPLE — NOT A MATCH**A****Asha**

Teaches Python
Wants Guitar

≠**C****Cara**

Teaches Design
Wants Python

Needs a 3-person loop to close.

03 / SOLUTION

Solution

One place to publish, discover, agree and record.

- One account acts as both learner and teacher.
- Publish skill: title (5–90), category (incl Other), outcome (10–300), tags 1–8, duration 15–120 min, online/in-person, proof link.
- Discover: search title/description/tag/@username; filters by category/language/offer/delivery; saved skills; learning-request posts.
- Arrange: free session · direct swap (2 separate times) · paid mentorship (demo credits, 100 start, 15-min hold, no real money).
- Agree: request → accept / decline / counter time → scheduled; conflict checks; reschedule before attendance.
- Learn: chat (polling), external meeting link, attendance (both confirm), work (10–2000) + feedback, reviews (1–5).
- Motivate: 12 badges + XP/level + leaderboard (weekly / monthly / last 7 days).

FLOW

Publish → Discover → Request → Agree → Learn → Record → Badge

Free

Swap

Paid*

* demo credits — not real money

Asha → **Cara** → **Ben** → **Asha**
Python Design Guitar

Trade Circle = 3 pairwise lessons, not 1 group call.

04 / KEY FEATURES

Key features

What actually works today — points only.

01

Account & profile

- 18+ check; password 10–128; demo 6–digit code (5 min, on–screen, no email/SMS)
- Profile: avatar 512KB, username 3–24 (unique per workspace), languages, timezone, wants 1–10, social links
- Isolated workspaces; persona switch; 1–hour invite link; logout

02

Skill management

- Create / edit / pause / reuse; Other category; tag chips (Enter/+), 1–8 unique
- Max 20 per user; soft delete (history stays); blocked if pending/scheduled/circle open

03

Discovery

- Search skill + people (@username); filters: category/language/offer/delivery
- Tabs: Discover / Skill trades / Free sessions; learning–request posts; saved skills

04

Bookings

- Incoming / sent / upcoming / completed / closed; counter–time; reschedule (before attendance)
- Attendance: both confirm; swap needs both lessons confirmed

05

Trade Circles

- Rule–based: 3 distinct, online, same language, exact tag↔want, no blocked pairs
- 1/3 → 2/3 → 3/3 consent; 3 bookings only then; withdraw cancels remaining

06

Chat & records

- Message request (1 intro) → accept; polling 3s; 500 msgs/convo, 100 loaded; unread counts
- Meeting link (teacher), evidence + feedback (10–2000), reviews (5–700, 1 per booking)

07

Badges (NEW)

- 12 badges: common/rare/epic; how to earn shown (e.g., 3 teaches, 1 circle, 5★)
- XP = sum; Level = floor(XP/200)+1; progress bar on profile + badges page

08

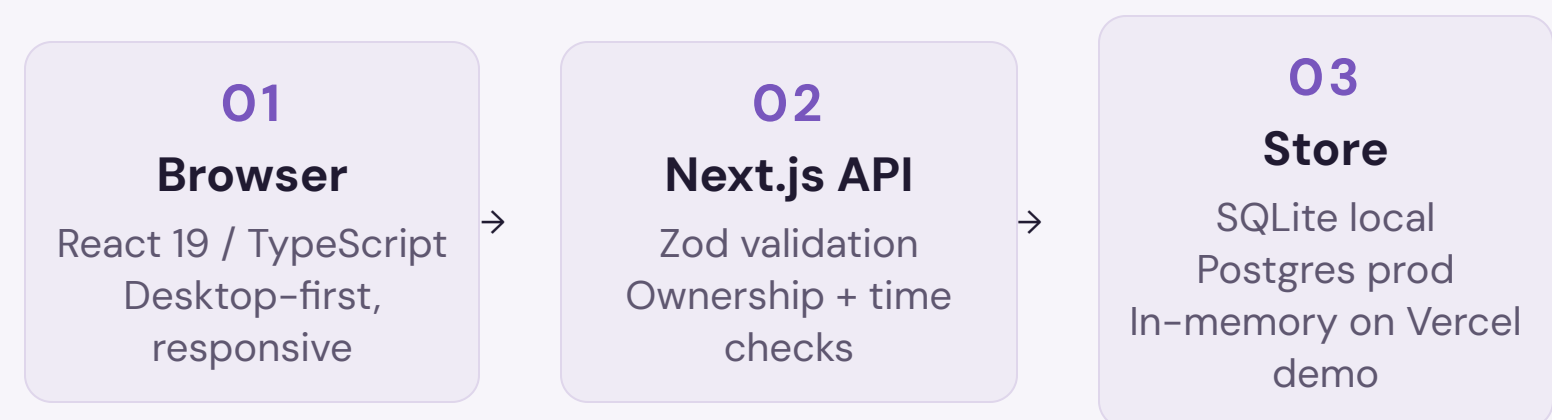
Leaderboard (NEW)

- Weekly (Mon–Sun UTC) / Monthly / Last 7 days (rolling)
- Points: +40 completed (+15 trade +20 circle) +25 skill +50 5★ +2 msg cap20 +15 post; podium + table

05 / HOW WE DOING IT · TERMINOLOGY

How we built it & what terms mean

How we do it



- One project: frontend + API together; transactions protect bookings/credits/messages.
- Polling: state 5s, chat 3s; not WebSocket; no end-to-end encryption.
- Limits: posts 150, circles 30, bookings 250 per workspace; build + typecheck + tests pass.

Terminology — points

Skill listing

Category, level, language, duration 15–120, delivery, tags, outcome, proof link.

Booking

States: pending → countered → awaiting-payment (15m) → scheduled → completed / declined / cancelled / expired.

Trade Circle

3 people, 3 skills, 3 pairwise lessons; exact match; 3/3 consent.

Badge

12 earned by action; tier common/rare/epic; XP 40–250.

XP / Level

XP = sum badge XP; Level = $\text{floor}(\text{XP}/200)+1$; XP forever, points per window.

Leaderboard

Weekly / Monthly / Last 7 days; points as above; sorted points → XP → name.

Workspace

Isolated seeded demo; shared via 1-hour invite; not global.

Credits

Fictional integer (100 = 1 credit); no purchase/withdrawal; refunds if balance allows.

06 / INNOVATION & IMPACT

Innovation & impact

Small wins, not hype — points only, site-only.

Innovation — points

- Goes beyond perfect 2-person match via 3-person Trade Circles (explainable tag match).
- Unanimous consent: lessons activate only at 3/3; time-conflict checks.
- Proof beyond attendance: separate evidence (learner) + feedback (teacher) → reviews.
- Motivation without money: 12 badges with clear how-to-earn + live leaderboard windows.
- One codebase for UI + API; seeded workspaces for parallel demo without real data.

Impact & benefits — points

- Learner: find exact outcome/ language; pay with knowledge when possible.
- Teacher: share one skill to many; visible minutes + reviews + badges.
- Community: impossible swaps become possible via loops; chat tied to learning.
- Safe demo: isolated workspaces, fictional contacts/credits, help & safety guidance.
- Honest limits: demo recognition, not accredited cert; no real payments; no staffed moderation; workspace expiry.

Share what you know · Learn what you need · Keep a record